

ASSIGNMENT 2: Linear Algebra for Machine Learning

These questions are designed to help you realize why Linear Algebra is the foundation of Machine Learning. Focus on understanding the intuition, not just formulas.

Section A: MCQs (Application-Oriented)

Question 1

In Machine Learning, a dataset is usually represented as:

- a) A single number b) A matrix c) A string d) A graph

Answer: b) A matrix

Datasets are conventionally stored with samples as rows and features as columns.

Question 2

Each row in a dataset matrix typically represents:

- a) A feature b) A model c) A data sample d) A loss function

Answer: c) A data sample

A row contains the measurements belonging to one observation.

Question 3

Each column in a dataset corresponds to:

- a) A prediction b) A feature c) A label d) A bias

Answer: b) A feature

A column stores one feature across the dataset samples.

Question 4

Why is matrix multiplication important in ML models?

- a) To sort data b) To transform inputs into outputs c) To store data d) To delete noise

Answer: b) To transform inputs into outputs

Multiplication by a weight matrix maps an input representation to an output representation.

Question 5

In a linear model ($y = Wx + b$), W represents:

- a) Output b) Input c) Weights d) Error

Answer: c) Weights

W contains the learned weights applied to the input.

Question 6

The equation $y = Wx + b$ is an example of:

- a) Matrix transformation b) Sorting algorithm c) Graph traversal d) Clustering

Answer: a) Matrix transformation

The expression applies a linear transformation followed by a bias shift.

Question 7

Feature scaling can be seen as:

- a) Matrix inversion b) Vector transformation c) Determinant calculation d) Rank reduction

Answer: b) Vector transformation

Scaling changes the coordinates or magnitude of a feature vector.

Question 8

Why are vectors used in ML?

- a) To store text b) To represent data numerically c) To compress images d) To delete features

Answer: b) To represent data numerically

Vectors encode features and other data as numerical coordinates.

Question 9

Dot product between two vectors measures:

- a) Distance b) Angle/similarity c) Rank d) Determinant

Answer: b) Angle/similarity

The dot product reflects alignment; with vector norms it gives cosine similarity.

Question 10

Cosine similarity is based on:

- a) Matrix inverse b) Dot product c) Rank d) Trace

Answer: b) Dot product

Cosine similarity is the dot product divided by the product of vector magnitudes.

Question 11

If two feature vectors are orthogonal, it means:

- a) They are identical b) They are independent c) They are zero d) They are scaled

Answer: b) They are independent

For nonzero vectors, orthogonality means zero dot product and linear independence.

Question 12

Why is linear independence important in ML?

- a) Reduces redundancy in features b) Increases data size c) Slows computation d) Removes labels

Answer: a) Reduces redundancy in features

Independent features avoid linear redundancy and improve identifiability.

Question 13

If features are linearly dependent, the model may:

- a) Improve accuracy b) Overfit or become unstable c) Run faster d) Remove noise automatically

Answer: b) Overfit or become unstable

Redundant predictors can make fitted weights unstable and contribute to overfitting.

Question 14

Rank of a dataset matrix helps determine:

- a) Number of outputs b) Effective information content c) Learning rate d) Loss value

Answer: b) Effective information content

Rank measures the number of independent directions represented in the data.

Question 15

Dimensionality reduction techniques aim to:

- a) Increase rank b) Reduce redundant features c) Add noise d) Increase variables

Answer: b) Reduce redundant features

These methods discard redundant dimensions while retaining useful structure.

Question 16

Principal Component Analysis (PCA) is based on:

- a) Sorting b) Eigenvalues and eigenvectors c) Graphs d) Strings

Answer: b) Eigenvalues and eigenvectors

PCA finds principal directions through eigenvectors of a covariance-related matrix.

Question 17

Eigenvectors in PCA represent:

- a) Noise b) Important directions in data c) Errors d) Labels

Answer: b) Important directions in data

Principal eigenvectors identify directions along which the data varies.

Question 18

Eigenvalues indicate:

- a) Data size
- b) Importance (variance) along a direction
- c) Number of rows
- d) Rank only

Answer: b) Importance (variance) along a direction

An eigenvalue gives the variance captured along its corresponding principal direction.

Question 19

A higher eigenvalue in PCA means:

- a) Less important feature
- b) More important direction
- c) Noise
- d) Zero variance

Answer: b) More important direction

A larger eigenvalue means that direction explains more variance.

Question 20

Why do we reduce dimensions in ML?

- a) To confuse model
- b) To remove redundancy and improve efficiency
- c) To increase complexity
- d) To remove labels

Answer: b) To remove redundancy and improve efficiency

Fewer dimensions can reduce redundancy, computation, and overfitting.

Question 21

In neural networks, weights are updated using:

- a) Scalars
- b) Matrices and vectors
- c) Strings
- d) Graphs

Answer: b) Matrices and vectors

Layer weights are arrays of learned numerical parameters, commonly matrices and vectors.

Question 22

Forward propagation involves:

- a) Matrix multiplication
- b) Sorting
- c) Searching
- d) Hashing

Answer: a) Matrix multiplication

A neural-network layer commonly computes a matrix-vector or matrix-matrix product.

Question 23

Backpropagation heavily uses:

- a) Determinants
- b) Gradients (vector calculus)
- c) Sorting
- d) Graph traversal

Answer: b) Gradients (vector calculus)

Backpropagation applies derivatives and propagates gradients through the network.

Question 24

Gradient descent updates weights using:

- a) Addition only
- b) Vector operations
- c) Strings
- d) Determinants

Answer: b) Vector operations

The weight vector or matrix is adjusted using gradient-based vector operations.

Question 25

Loss functions are minimized using:

- a) Matrix transformations
- b) Vector optimization
- c) Sorting
- d) Searching

Answer: b) Vector optimization

Optimization searches parameter space for values that reduce the loss.

Question 26

Why is matrix inversion sometimes needed in ML?

- a) To remove data
- b) To solve linear equations (e.g., regression)
- c) To scale features
- d) To visualize data

Answer: b) To solve linear equations (e.g., regression)

Linear regression can be formulated as a system or least-squares problem; inversion is one possible solution method.

Question 27

Overcomplete feature sets lead to:

- a) Higher rank b) Redundancy c) Faster models d) Perfect accuracy

Answer: b) Redundancy

More features than independent data directions can introduce redundant representations.

Question 28

If rank < number of features, it indicates:

- a) All features useful b) Redundant features exist c) No data d) Perfect dataset

Answer: b) Redundant features exist

Rank below the number of feature columns means those columns are linearly dependent.

Question 29

Singular matrices in ML indicate:

- a) Perfect data b) Redundancy or dependency c) High accuracy d) No features

Answer: b) Redundancy or dependency

A singular feature or design matrix often signals linearly dependent columns.

Question 30

Why do we use normalization?

- a) To reduce matrix size b) To stabilize numerical computations c) To delete features d) To increase rank

Answer: b) To stabilize numerical computations

Scaling inputs to comparable ranges can improve numerical behavior and optimization stability.

Question 31

Feature vectors in ML live in:

- a) 1D space b) n-dimensional space c) Graph space d) Tree space

Answer: b) n-dimensional space

A feature vector with n components is a point in n-dimensional space.

Question 32

Distance metrics (like Euclidean) are based on:

- a) Linear algebra b) Sorting c) Graphs d) Strings

Answer: a) Linear algebra

Euclidean distance is calculated from differences between vector coordinates.

Question 33

Clustering algorithms rely on:

- a) Distance between vectors b) Sorting c) Searching d) Hashing

Answer: a) Distance between vectors

Many clustering methods group samples according to distances between their feature vectors.

Question 34

Why is orthogonality useful?

- a) Features become independent b) Features become identical c) Rank becomes zero d) Noise increases

Answer: a) Features become independent

Orthogonal directions have zero dot product and no linear overlap; this is not, in general, the same as statistical independence.

Question 35

In recommendation systems, similarity is computed using:

- a) Dot product / cosine similarity b) Sorting c) Determinant d) Rank

Answer: a) Dot product / cosine similarity

Recommendation systems often compare user or item embeddings with dot products or cosine similarity.

Question 36

In image processing, an image is treated as:

- a) Scalar b) Matrix c) Graph d) String

Answer: b) Matrix

A grayscale image is represented by a 2D matrix; a colour image is typically a higher-dimensional tensor.

Question 37

Compression techniques rely on:

- a) Rank reduction b) Sorting c) Searching d) Hashing

Answer: a) Rank reduction

Low-rank methods compress data by retaining fewer independent components.

Question 38

Low-rank approximation helps in:

- a) Increasing noise b) Reducing storage and computation c) Increasing dimensions d) Removing labels

Answer: b) Reducing storage and computation

Approximating with fewer components can reduce storage and computation.

Question 39

Why are tensors used in deep learning?

- a) To avoid matrices b) To generalize matrices to higher dimensions c) To reduce features d) To sort data

Answer: b) To generalize matrices to higher dimensions

Tensors extend vectors and matrices to arrays with more dimensions.

Question 40

Bias term in ML models helps:

- a) Reduce dimensions b) Shift decision boundary c) Remove noise d) Increase rank

Answer: b) Shift decision boundary

The bias adds an offset and can shift a model's decision boundary.

Question 41

A linear classifier creates:

- a) Curve b) Hyperplane c) Circle d) Random shape

Answer: b) Hyperplane

A linear classifier separates classes with a hyperplane.

Question 42

Hyperplanes are defined using:

- a) Vectors and dot products b) Sorting c) Strings d) Graphs

Answer: a) Vectors and dot products

A hyperplane can be expressed using a normal vector and a dot product.

Question 43

If two classes are linearly separable, it means:

- a) They overlap b) A line/plane can separate them c) No boundary exists d) Infinite solutions

Answer: b) A line/plane can separate them

Linear separability means a line, plane, or higher-dimensional hyperplane separates the classes.

Question 44

Kernel methods extend:

- a) Scalars b) Linear algebra to higher dimensions c) Sorting d) Searching

Answer: b) Linear algebra to higher dimensions

Kernel methods use inner products to model relationships in an implicit higher-dimensional feature space.

Question 45

Why is sparsity useful?

- a) Increases storage b) Reduces computation c) Adds noise d) Increases rank

Answer: b) Reduces computation

Sparse representations store or process fewer nonzero values, often reducing computation.

Question 46

Feature embeddings convert data into:

- a) Strings b) Vectors c) Graphs d) Scalars

Answer: b) Vectors

Embeddings map items such as words or entities to numerical vectors.

Question 47

Matrix factorization is used in:

- a) Sorting b) Recommendation systems c) Searching d) Hashing

Answer: b) Recommendation systems

Matrix factorization is widely used to infer user and item factors in recommendation systems.

Question 48

If two vectors are very similar, their cosine similarity is:

- a) Near 0 b) Near 1 c) Negative d) Infinite

Answer: b) Near 1

Similar vectors pointing in nearly the same direction have cosine similarity close to 1.

Question 49

If vectors are opposite, cosine similarity is:

- a) 1 b) 0 c) -1 d) Infinite

Answer: c) -1

Opposite nonzero vectors have cosine similarity -1.

Question 50

The core idea behind ML models can be summarized as:

- a) Data + Linear Algebra + Optimization b) Sorting + Searching c) Graphs + Trees d) Strings + Parsing

Answer: a) Data + Linear Algebra + Optimization

Machine learning combines data representations, mathematical models, and optimization.